

A 1Mb RRAM Macro with Bipolar Forming for Improved Programming Yield and Cell-by-cell Write Verification Scheme.

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Abstract— Resistive RAM (RRAM) has emerged as a promising candidate for the next generation non-volatile memories (NVMs) due to its low write voltage, compact area, and CMOS compatibility. In this work, we propose a 1Mb RRAM macro with bipolar forming to reduce the forming voltage and improve the programming yield. Additionally, a cell-by-cell write verification scheme is introduced to protect RRAM cells from overstress and improve RRAM yield. The test chip, fabricated using 40nm CMOS technology, occupies a core area of 2.34 mm².

Keywords— RRAM, non-volatile memory, Unipolar forming, Bipolar forming, Cell-by-cell write verification.

I. INTRODUCTION

Flash memory has been the leading non-volatile memory (NVM) technology due to its high-density data storage capabilities, maintaining its dominance at least through the 2X-nm generation. [1-2]. However, the intrinsic physical characteristics of flash memory face significant limitations beyond certain technical nodes, such as high leakage current and slow write speed. As a result, novel NVM technologies are being developed to achieve high density, low power, and fast operation. Several emerging NVMs such as magnetoresistive random-access memory (MRAM) [4-5], ferroelectric RAM (FeRAM) [6], phase-change RAM (PCRAM) [7-8], and resistive RAM (RRAM) [9-12] are introduced as promising alternatives due to improved write speed and reduced power consumption compared to embedded flash memory. However, MRAM has limited density, FeRAM faces challenges with scalability, while PCRAM suffers from high energy consumption during write operation. RRAM is considered an excellent NVM among these emerging NVM technologies due to its advantageous characteristics, such as low operating voltage (<3V), fast switching speed (<10ns), compact area (<10nm), large resistance ratio (10~1000), and excellent compatibility with CMOS technology [2-3]. These advantages of RRAM make it appealing for next-generation NVM memory in high-density and high-speed NVM applications.

The 1T1R logic process-compatible RRAMs in [13-17] demonstrated strong suitability for embedded NVM applications that operate at low power and require fast read, particularly for energy-constrained devices. Our prior work in [25] demonstrated a low-power 1Mb RRAM macro with reliable sensing by a two-stage voltage-mode sense amplifier (VSA) and improved read speed. In this work, we proposed an improved 1Mb RRAM macro that employs dual-mode write drivers to support unipolar and bipolar forming. A write verification scheme is also employed to improve the write yield of the developed RRAM.

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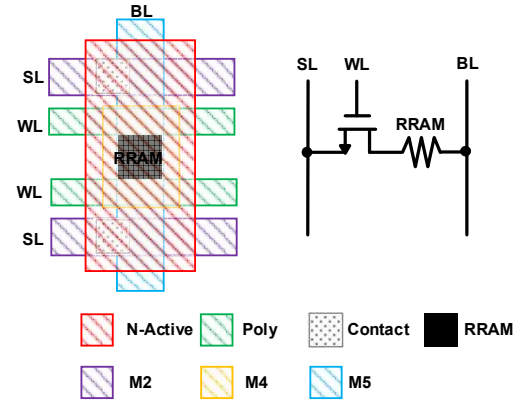


Fig. 1 RRAM bitcell layout (left) and schematic (right).

II. BACKGROUND

A. 1T1R RRAM Bitcell and Basic Operations

RRAM is a metal-insulator-metal (MIM) structure that can switch between a high-resistance state (HRS) and a low-resistance state (LRS), depending on the direction of the current flowing through it. The 1T1R bitcell consists of an NMOS transistor and an RRAM connected in series, as shown in Fig. 1. In the configuration, the access transistor selects and controls the RRAM by enabling the word-line (WL), while the RRAM works as a storage device by switching the state between HRS (“0”) and LRS (“1”). To realize column by column programming access and cell-by cell write verification, we have implemented a split gate and source-line (SL) design, resulting in bitcell area of 0.135 μm^2 .

Fig. 2 illustrates the fundamental operations performed on the RRAM array. Initially, fresh cells require once forming operation to induce soft breakdown of the dielectric by

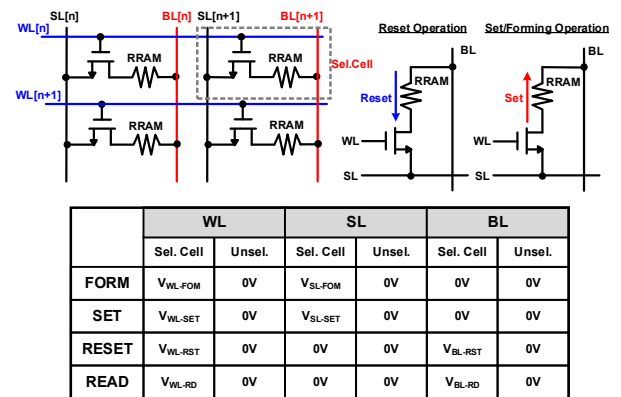


Fig. 2 RRAM array structure and basic operations.

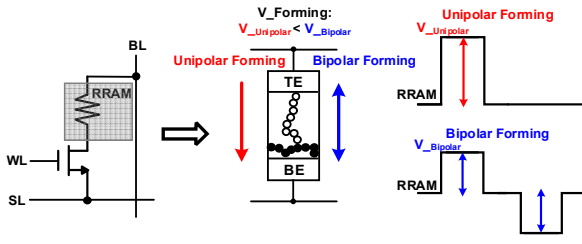


Fig. 3 Simplified operation of unipolar forming and bipolar forming.

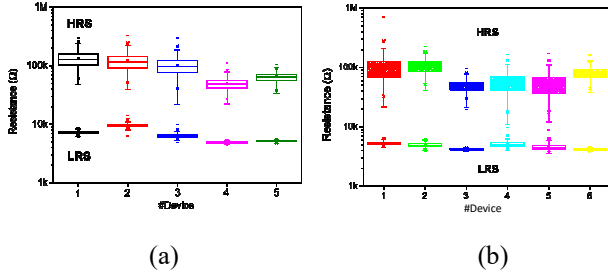


Fig. 4 measurement result of (a) unipolar forming and (b) bipolar forming.

applying high forming voltages (V_{WL-FOM} and V_{SL-FOM}) to the WL and the SL, while grounding the BL before programming. It is a one-time process to switch the RRAM devices from initial HRS to LRS. However, unselected cells in same row have 0 V on both BL and SL. This ensures that there is no voltage difference across the unselected cells, which prevents unwanted forming operations. After the forming operation, RRAM cells are ready for programming. “SET” refers to the RRAM cell transition from HRS to LRS, corresponding to writing “1.” It is similar to the forming operation. However, the applied bias voltage is lower than the forming voltage. The opposite process, known as “RESET”, switches the selected RRAM cell from LRS to HRS by applying a write voltage to the BL. Simultaneously, the BLs and SLs of unselected cells are grounded, as shown in the table of Fig 2. During the read operation, the BL is precharged with a low voltage to avoid unwanted partial resetting of the RRAM and state flipping.

B. Motivation for Realizing Bipolar Forming Operation

In RRAM operations, forming is an essential process for creating conductive filaments to activate resistive switching characteristics. Typically, the forming voltage tends to be higher than the SET (RESET) voltage, which switches the

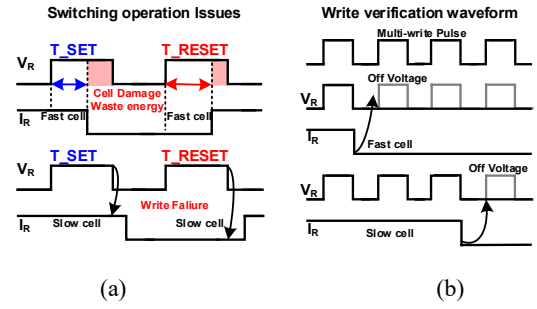
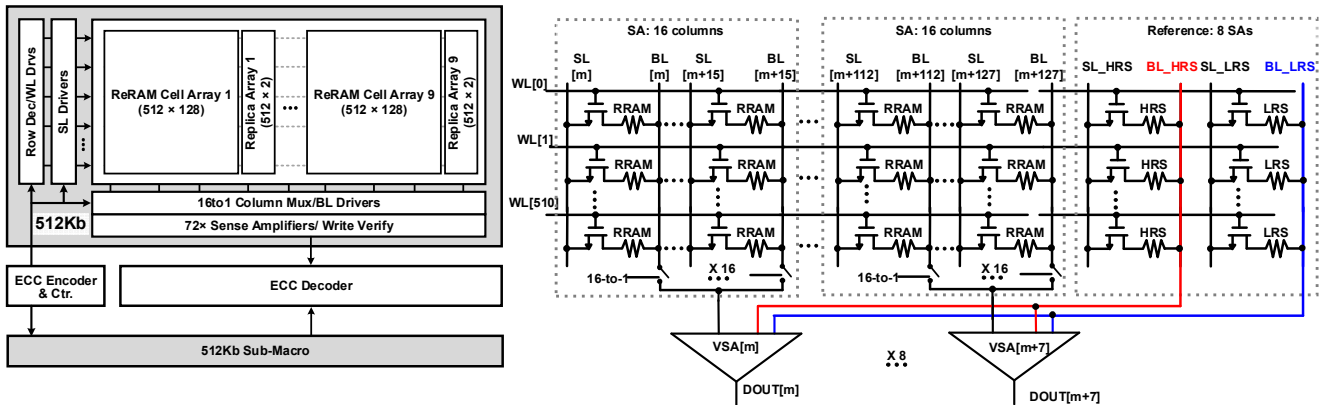


Fig. 5 Waveform of (a) switching issues and of (b) write verification operation.

device from HRS (LRS) to LRS (HRS). As the device size decreases, reducing the forming voltage becomes a critical challenge. Unipolar forming has been the mainstream forming methodology by applying a single pulse on one side of an RRAM device. As shown in Fig. 3, bipolar forming by applying bipolar polarity pulses on both sides of an RRAM device has been employed to reduce the forming voltage. The bipolar pulse can reduce forming voltage without affecting essential RRAM operation characteristics and degraded device performance while maintaining good reliability [18-19]. Fig. 4 shows the measurement results for unipolar forming with a forming voltage of 2.5 V and bipolar forming with forming voltages of 1.9 V and 1.8 V. These results demonstrate that bipolar forming requires a lower forming voltage. The developed write driver supports unipolar and bipolar forming.

C. Motivation of Write Verification Concept

During the write operation, reliable resistance switching is critical for the practical application of RRAMs. Conventionally, fixed amplitude and time width are used for programming. However, due to the randomness of the formation of conductive filaments, the switching process of RRAM also exhibits variability. Fast-switching cells suffer excessive energy consumption, long stress time, endurance degradation, and over-write issues as depicted in Fig.5(a) [20-21]. Meanwhile, for the slow cell, a fixed amplitude or time generates variations in the programmed resistance, leading to SET and RESET failures. These variations reduce the R-ratio (R_{HRS}/R_{LRS}) of RRAM and impact the retention and endurance cycles [20-24]. Therefore, to improve the write yield, write verification is required. Multiple writes may be applied to program the slow cell. In this operation, the write



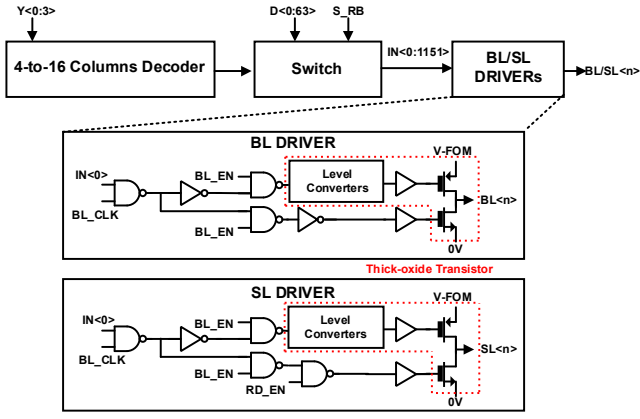


Fig. 7 Proposed BL/SL driver

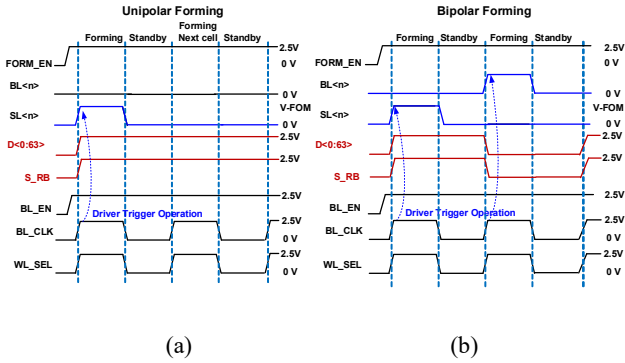


Fig. 8 Timing waveform of (a) unipolar forming and (b) bipolar verification process can check the status after the programming of the RRAM cell shown in Fig. 5(b) to disconnect the program voltage across the RRAM.

III. PROPOSED RRAM MACRO DESIGN

A. Overall Architecture

Fig. 6 (left) illustrates the overall architecture of the 1Mb RRAM macro, which consists of two 512Kb sub-macro, an error checking code (ECC) encoder and a decoder. Each 512Kb sub-macro consists of an RRAM subarray, a row decoder, a column decoder, WL/SL/BL drivers, $72 \times$ voltage-mode sense amplifiers (VSAs), and write verification circuits. The proposed 1Mb RRAM macro includes a reference array and a write verification block compared to the prior 1Mb macro. The RRAM macro also activates only one sub-macro for each operation to reduce unnecessary power consumption. Each RRAM sub-array consists of 512×1152 RRAM cells, with eighteen columns for generating the reference voltages, and 128 columns for storing the parity bits. In the prior RRAM macro, two replica columns are shared by 72 SAs. However, to reduce reference variation, 72 SAs share 18 replica columns, which are evenly distributed in the RRAM array (i.e., every two replica columns are located at the end of 128 RRAM columns). Fig. 6(right) demonstrates the diagram between the RRAM array and SAs. A 16-column RRAM array connects to the input of the SA by using a 16-to-1 multiplexer. Replica cells are shared among 8 SAs to generate the reference voltage. When writing data, a 16-to-1 multiplexer connects the BL and SL, allowing 72 data bits to be written simultaneously.

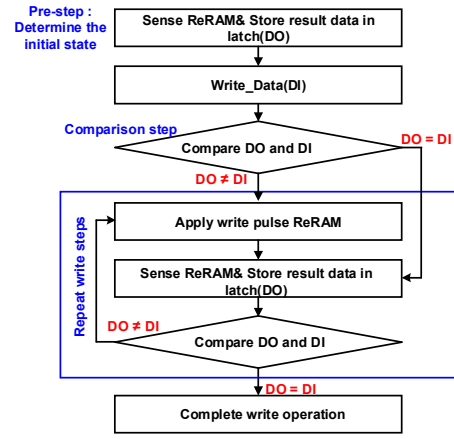
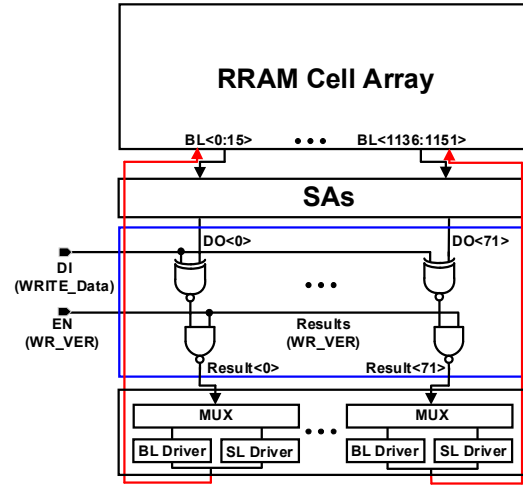


Fig. 9 Flowchart of write verification.



Comparison (DO and DI)	EN (WR_VER)	Result (WR_VER)	SL	BL
0(DO≠DI)	1	1	V_{SL-SET}	V_{SL-RST}
1(DO=DI)	1	0	0	0

Fig. 10 Write verification scheme and truth table.

B. Proposed Bipolar Forming Driver.

Fig. 7 shows the proposed BL/SL driver structure. The BL/SL drivers are controlled by BL_CLK and BL_EN. The switch block's D<0:63> and S_RB signals determine whether the forming voltage is applied to the BL or SL side. For example, when both D<0:63> and S_RB are high, the forming voltage is driven to the SL. Conversely, when both D<0:63> and S_RB are low, the forming voltage is driven to the BL. These signal operations support both unipolar and bipolar forming operations. Fig. 8 shows the waveform for unipolar and bipolar forming operations. The unipolar forming voltage is applied only to SL. However, in bipolar forming, both BL and SL have the forming voltages.

C. Proposed Write Verification.

The basic concept of write verification scheme is to compare RRAM data with external input data to determine whether to generate a write voltage pulse. Fig. 9 shows the flowchart. The write verification operation is divided in three

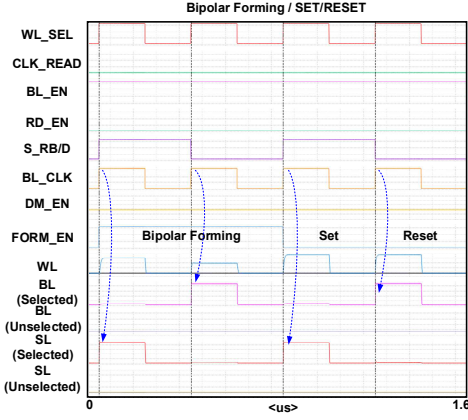


Fig. 11 Simulation waveform of bipolar forming pulse and write pulse.

steps: before the write verification step, the initial states are read, and the results are compared with the input data to decide whether to proceed with write verification. During write verification operations, writing, reading, and comparison are repeated until the input data (DI) and output data (DO) match. Fig. 10 illustrates the detailed write verification structure, which consists of 72 XNOR and 72 NAND. The stored DI and DO from the SA output are inputs of XNOR. In this stage, the values of DI and DO are compared in the XNOR. For example, if they are different, the result will be a 0 signal; if they are the same, the result will be a 1 signal. This result is used as the input of the next NAND stage with the WR_VER signal. The output result of this stage determines verification. When the verification condition is met, the output result of this stage disables the BL/SL voltage. Since this macro implements split SL, cell-by-cell write verification occurs simultaneously across 72 cells. For cells that do not meet the verification condition, the write operation is repeated until all cells satisfy the verification condition. In the case of fast-write cells where the verification condition has already been met, this can be maintained without damaging the cells.

IV. EVALUATION

Fig. 11 shows the transient BL and SL voltage, pulse for bipolar forming, SET, and RESET. This result is divided into two parts: Form_EN on and off. Bipolar forming is set by Form_EN, S_RB, and D. Forming voltage is applied to BL and SL according to WL_SEL and BL_CLK. Our implemented design adjusts WL voltage to reduce forming voltage across RRAM. When Form_EN turns off, the write driver applies programming states. Fig. 12 shows the programming results, including write verification. This result operates step by step, performing write and read operations

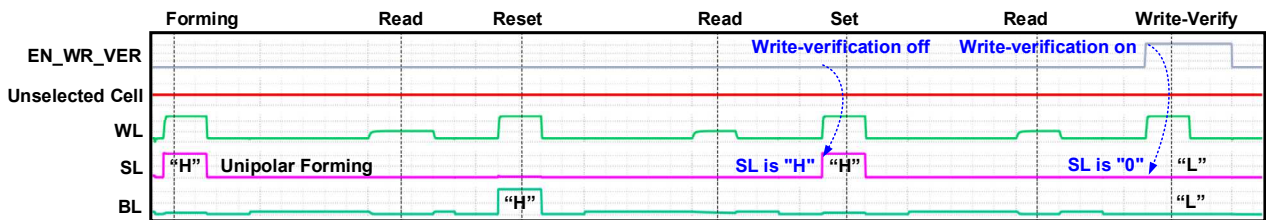


Fig. 12 Simulation waveform of programming and read including write verification.

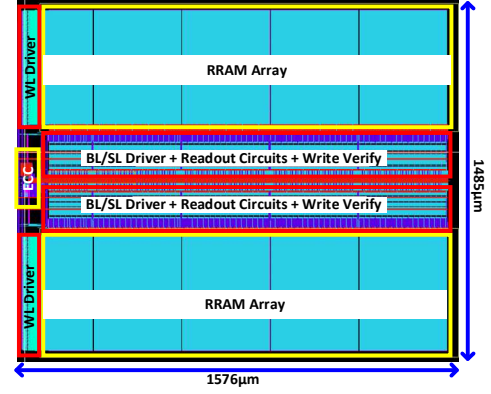
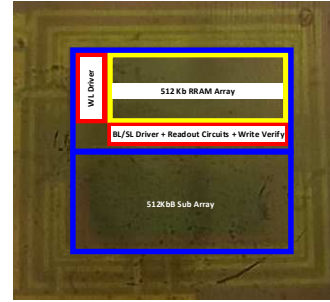


Fig. 13 Layout of the 1Mb RRAM macro.



Technology	40nm Process
Capacity	1Mb (2 × 512Kb)
Architecture	Split SL
Sub-Array (512Kb)	512 × 1024
Core Area	1576μm × 1485μm
Cell Size	0.41μm × 0.328μm
Key Feature	1: Unipolar / Bipolar Forming 2: Write Verification
Amplitude	WL: 1.2V- 2.5V BL: 1.2V-3.5V SL: 1.2V-3.5V

Fig. 14 Die photo and Summary table.

after unipolar forming. While the write verification operation is active, the SL voltage remains at 0 during the SET operation. This condition occurs when the SET operation follows the completion of the previous SET operation. As a result, programming does not occur in the RRAM. Fig. 13 and Fig. 14 show the layout, die photo of the 1Mb RRAM macro, and summary table. The 1Mb RRAM macro is comprised of two 512Kb sub-macros that are fabricated using 40nm technology. It includes a write driver that supports both unipolar and bipolar forming operations as well as a write verification scheme.

V. CONCLUSION

In this paper, a new 1Mb RRAM macro comprising two 512Kb sub-macros is proposed, which is fabricated using a 40nm process with logic-process-compatible RRAM devices. The new macro implements two key features: the first is a write driver that supports bipolar forming operation to reduce the forming voltage. The second is a write verification scheme to enhance write yield with a split SL, compared to the previous 1Mb macro [25].

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